

DAVID DUNN

Applied AI & Integration Engineer

Santa Clara, CA | US Citizen | (305) 833-3909 | davidalexanderdunn@gmail.com
daviddunn.dev | github.com/algorithmic | linkedin.com/in/mrdaviddunn

SUMMARY

Applied AI & Integration Engineer with 6+ years building data pipelines, BI platforms, automation workflows, and LLM-powered systems. Experienced connecting APIs, databases, dashboards, AI models, infrastructure, and human-in-the-loop workflows into production-ready tools for analytics, decision support, and operational automation. Recent work includes agent orchestration, real-time anomaly detection, AI-assisted onboarding, semantic clustering, and full-stack workflow platforms.

TECHNICAL SKILLS

Applied AI & LLMs: Claude, OpenAI, Vertex AI, AI agents, LLM workflows, semantic clustering, prompt engineering, MCP servers

Integration & Automation: API integration, workflow automation, Airflow, WebSockets, event-driven systems, Docker, Git, Linux

Data Engineering: Python, SQL, dbt, Kafka, Spark, DuckDB, PostgreSQL, SQLite, Snowflake, Redshift, BigQuery

Analytics & ML: anomaly detection, A/B testing, forecasting, statistical modeling, scikit-learn, statsmodels

BI & Product: Metabase, Looker, Tableau, Power BI, Streamlit, dashboard development, stakeholder requirements

TECHNICAL PROJECTS

Saga - Multi-Agent Orchestration System | Docker · MQTT · Gitea · Claude Code · Telegram · Proxmox

- Built a multi-agent orchestration system on Docker/MQTT/Gitea that runs parallel Claude Code workers against Telegram-based spec intake, with human review gates and self-healing PR loops for long-running projects.
- Codified agent roles, message-passing protocols, and review artifacts into reusable deployment patterns for autonomous coding workflows in production.

Nexus - Real-Time Prediction Market Intelligence Engine | Python · WebSockets · DuckDB · Claude · Convex

- Built real-time prediction market intelligence engine ingesting 100K+ events with WebSocket streaming, integrity validation, and async event-driven architecture.
- Developed Claude-powered semantic market clustering, sliding-window anomaly detection, and cross-market correlation; backed by an evaluation framework of 158 tests across 17 modules.

marketfinder - Live Dashboard for Nexus | React · Vite · TypeScript · Convex · Tailwind

- Built the user-facing intelligence layer for Nexus, turning 10K+ live prediction markets into searchable dashboards, anomaly alerts, trend clusters, and comparison views.
- Shipped a responsive analytics interface using Convex live queries, TanStack React Table, severity filters, and mobile-first card layouts.

PROFESSIONAL EXPERIENCE

Applied AI & Integration Engineer | Client Projects | Santa Clara, CA

10/2025 - Present

- Built [Happily.love](https://happily.love) end-to-end for a client: Claude-assisted onboarding interviews, invite-based applicant flow, structured profile model, approval workflows, and private matchmaker dashboard.
- Designed human-in-the-loop decision-support tooling that converts qualitative interviews and matchmaker notes into structured candidate profiles for review, search, and matching.

Data & BI Solutions Developer | Dunn & Done Systems · Remote | Santa Clara, CA

10//2024 - 10/2025

- Built reusable Python/Airflow ELT workflows for SMB clients, normalizing CRM, payment, ad-platform, and accounting data into AWS-backed warehouse models with dbt lineage and quality checks; cut recurring reporting effort 75%.
- Developed LLM-assisted validation and metric-reporting workflows on top of S3, Redshift, dbt, and Metabase, checking metric definitions, flagging data quality issues, and supporting revenue, churn, and LTV reporting for SMB clients.

Analytics Engineer | Front Row | Remote

12/2022 - 09/2024

- Authored modular dbt models and Looker/Metabase explores answering 80% of recurring product-marketing questions, reducing ad-hoc requests 60%; owned metric definitions and quality rules.
- Built an experimentation pipeline using GTM, feature flags, dbt, and Python uplift analysis to support 20+ A/B tests and dynamic-pricing tactics that lifted average purchase price 8%.

Marketing Data Analyst | ORComm | Seoul, South Korea

01/2020 - 08/2022

- Created reusable marketing analytics dashboards for 10 seed-stage consumer startups, standardizing CAC, ROAS, cohort retention, and campaign performance reporting; reduced recurring report build time 60%.
- Created cohort, funnel, and retention analyses to identify high-risk user segments, improve lifecycle targeting, and standardize metric definitions across client reporting.

Business Intelligence Analyst | VR Comet | Seoul, South Korea

12/2018 - 12/2019

- Transformed Google Analytics, HubSpot, and Unity event data in Redshift into reusable SQL views and lightweight reports for retention, monetization, and acquisition analysis, helping identify growth opportunities tied to 15% MoM DAU growth.

EDUCATION

B.S. Physics - Lake Forest College - 2009 | Data Analyst in R - [Dataquest.io](https://dataquest.io) - 2022 | Data Scientist in Python - DataCamp - 2024